
The Determinants of Firm Exporting in Argentina

Abstract: This study analyzes the determinants of Argentine firms' participation in international trade in 2017. Argentina is an emerging economy characterized by recurrent macroeconomic instability. The capacity of enterprises to export is thus of critical importance for the country's economic stability. The paper examines how firms' structural, organizational, technological, and geographic characteristics influence their export propensity. The empirical analysis draws upon data from the World Bank Enterprise Survey (WBES) conducted in 2017 across approximately 1,000 Argentine firms. The objective is to identify and quantify the influence of various factors on firms' participation in foreign markets using statistical and econometric techniques. The findings indicate that firm size, membership in the manufacturing sector, and geographical location significantly increase export probability, whereas logistical constraints, lower workforce qualification, and perceived political instability are associated with diminished engagement in international trade.

Group 29

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1 Introduction

In an era of heightened globalization, a country's export performance serves as a primary indicator of its economic competitiveness and degree of integration into global trade networks. Exports play a vital role in national economic expansion: they generate foreign currency receipts, bolster the balance of payments equilibrium, and broaden corporate sales outlets beyond domestic boundaries. They also contribute fundamentally to the trade balance, defined as the net difference between export and import values. A robust export volume thereby enables the financing of capital goods and intermediate imports essential to productive activity while mitigating reliance on foreign borrowing. At the macroeconomic level, a dynamic export sector fosters economic stability, enhances the cross-sector allocation of productive resources, and stimulates aggregate productivity.

Nevertheless, international economics literature demonstrates that firms participate unevenly in international trade. Seminal contributions by Bernard and Jensen (1999) alongside the heterogeneous firm framework formulated by Melitz (2003) emphasize that enterprises display substantial productivity differentials. Within this framework, only sufficiently productive firms can cover the sunk entry costs associated with accessing foreign markets. These costs generate an entry barrier that induces self-selection: less productive firms remain restricted to the domestic market, whereas their more productive counterparts expand internationally.

Whenever an enterprise succeeds in penetrating foreign markets, international integration can generate substantial economic gains. Access to broader consumer demand allows enterprises to capture economies of scale by expanding output volume and driving down unit costs. Furthermore, foreign trade exposure may stimulate learning mechanisms, widely conceptualized in the literature as "learning by exporting". Direct contact with foreign competition and higher quality standards incentivizes companies to optimize production methods, adopt advanced technologies, and elevate output quality. Consequently, exporting firms systematically display superior productivity levels, more structured operational frameworks, and stronger innovative capabilities.

Beyond direct growth impulses, exports contribute substantially to structural economic transformation. International openness encourages productive specialization aligned with comparative advantages. Standard trade theory suggests that economies gain by concentrating resources in lines of production where they possess relative cost or productivity efficiencies. Such specialization drives overall efficiency and optimizes resource allocation across sectors.

Furthermore, engaging in global trade fosters technological diffusion. Export-oriented enterprises frequently encounter stringent technical requirements and sophisticated competition abroad. Such pressures encourage ongoing investments in research and development, process optimization, and technological modernization. Trade openness can thus act as a catalyst for innovation and technical modernization throughout the domestic productive fabric.

In this light, empirical investigation into the microeconomic determinants of export activity proves crucial. Identifying which enterprises succeed in exporting and uncovering the underlying mechanisms clarifies the structural catalysts that support integration into international markets. These determinants encompass firm-internal factors—such as size, productivity, human capital, and technological endowments—as well as external variables including infrastructural quality, institutional environments, and logistical access to world markets.

These mechanisms assume particular salience in Argentina. The Argentine economy is persistently constrained by a foreign exchange bottleneck, marked by recurrent difficulties in generating sufficient foreign currency (notably US dollars) to meet its import needs. The country relies heavily on imported intermediate inputs, capital goods, and energy to sustain industrial activity. Yet its exports remain concentrated in a narrow basket of primary and agro-industrial commodities, such as soybeans, cereals, and meat products. When export revenues fail to offset import expenditures, the nation must resort to external debt or attract foreign capital inflows to balance its external

accounts. Hence, sustained foreign currency generation through goods exports represents an indispensable pillar of macroeconomic stability.

This structural balance-of-payments vulnerability reflects a distinct historical path. During the early twentieth century, Argentina ranked among the world's most prosperous nations, boasting per capita GDP levels on par with leading Western European economies like Germany. However, from the mid-twentieth century onward, the country underwent alternating shifts in economic strategy, swinging between phases of trade liberalization—characterized by tariff reductions and international market alignment—and inward-looking protectionist regimes marked by heightened tariffs, foreign exchange rationing, and extensive state intervention. These recurrent policy reversals generated persistent macroeconomic instability, leaving a lasting negative imprint on economic performance.

Among the most salient expressions of this structural volatility is chronically high inflation. Argentina experienced episodes of hyperinflation during the 1980s, with price increases exceeding 100% in the late 1970s and reaching over 3,000% in 1989. Despite multiple stabilization plans, inflation remained elevated across subsequent decades. It hovered near 25% in 2015, reached roughly 40% in 2016, and climbed past 50% toward the close of the 2010s. More recently, inflationary pressures surged dramatically, with annual inflation exceeding 200% by 2023. This dynamic is deeply rooted in structural fiscal deficits frequently financed via central bank monetization, compounded by limited policy credibility.

Economic policies pursued throughout the 2000s and 2010s also profoundly shaped the operational environment of Argentine firms. Under the administrations of Néstor Kirchner and Cristina Fernández de Kirchner, authorities implemented rigorous capital and exchange controls, commonly termed the *cepo cambiario*, coupled with heavy taxes on agricultural and primary exports (*retenciones*). Although designed to capture fiscal revenue and protect domestic purchasing power, these measures constrained firms' access to foreign inputs, restricted capital repatriation, and curtailed the price competitiveness of key exporting sectors.

The election of Mauricio Macri in late 2015 heralded an economic policy shift toward trade and financial liberalization. The new administration rolled out reforms including the dismantling of exchange controls and selective reductions in export tariffs. These adjustments coincided with a substantial devaluation of the Argentine peso intended to restore trade competitiveness. In theoretical terms, exchange rate depreciations can stimulate outward sales by improving relative price competitiveness, contingent upon the Marshall-Lerner elasticity condition.

More recently, the election of Javier Milei in late 2023 took place amidst severe economic turmoil characterized by triple-digit inflation and extensive currency depreciation. The administration launched an aggressive stabilization and deregulation agenda centered on rapid fiscal consolidation, deep cuts to public expenditure, subsidy removals, and market deregulation. These interventions aim to eradicate central bank deficit financing, widely viewed as the primary driver of domestic inflation. Early indicators throughout 2024 pointed toward gradual inflation deceleration and a rare primary budget surplus. While controversial due to short-term socioeconomic costs, these measures aim to rebuild institutional credibility and stabilize macroeconomic conditions. Over time, monetary and exchange rate stability could enhance investment visibility and strengthen firms' capacity to pursue long-term export strategies.

Alongside domestic macroeconomic dynamics, regional trade integration represents an essential determinant of Argentine export prospects. Argentina is a founding member of Mercosur, a customs union established in 1991 with Brazil, Uruguay, and Paraguay. By dismantling internal tariffs and adopting a Common External Tariff (CET), Mercosur was conceived to foster South American economic and commercial integration. This regional framework stimulated reciprocal trade flows by facilitating goods transit and enlarging the immediate accessible market. For Argentine companies, regional integration provides a crucial market outlet, particularly toward Brazil, which remains one of the country's preeminent commercial partners.

Regional integration within Mercosur has also fostered regional production networks, notably across automotive

manufacturing and agro-industrial processing. Firms can divide distinct stages of manufacturing across member economies, reinforcing intra-bloc value chains. Furthermore, access to a wider common market enables firms to capture scale economies, expanding production runs and driving down unit overhead.

Nevertheless, regional integration presents complex trade-offs. Elevated competition among member-state producers can strain less competitive domestic sectors. In Argentina's case, productivity gaps with neighboring economies, most notably Brazil's industrial core, can exacerbate bilateral trade imbalances and leave sensitive manufacturing branches vulnerable to regional competition.

Against this backdrop of severe macroeconomic challenges alongside regional trade opportunities, the year 2017 offers a particularly informative empirical setting. It captures a period of relative trade openness following initial reforms under the Macri administration, while remaining exposed to underlying macroeconomic fragilities.

Analyzing why specific enterprises successfully export while others remain confined to the domestic market is therefore of profound academic and policy relevance. Investigating the microeconomic determinants of export propensity identifies the firm-level characteristics that foster internationalization and clarifies the primary operational bottlenecks firms face.

This investigation utilizes firm-level data from the 2017 World Bank Enterprise Survey (WBES), which provides detailed information regarding enterprise structure, productivity, operational barriers, and global market participation. Accordingly, this paper addresses the following research question:

What are the determinants of firm-level exporting in Argentina in 2017?

The paper proceeds as follows. Section 2 reviews the theoretical and empirical literature to establish the analytical framework and outline hypotheses. Section 3 outlines the dataset, data hygiene procedures, and descriptive statistics. Section 4 presents the econometric framework, and Section 5 interprets the estimation results. Section 6 develops predictive profiling, followed by public policy recommendations in Section 7 and concluding remarks in Section 8.

2 Literature Review

The literature review explores the determinants of exporting by elucidating the full spectrum of factors influencing firms' capacity to participate in international markets. It demonstrates that export performance stems from a combination of internal factors related to firm-specific characteristics, such as size, human capital, and industrial sector. To these are added external factors, notably the geographic distribution of exports and regional disparities in infrastructure and logistical access. Finally, it highlights the importance of investment factors, such as technological initiatives, research and development, and international quality certifications, which govern product upgrading and foreign trade competitiveness.

2.1 Internal Factors

Internal factors constitute a foundational analytical framework for understanding export behavior. They encompass firm size, workforce quality, technological capacity, innovation orientation, and baseline productivity.

Firm Size

Firm size exerts a powerful influence on export engagement. Broadly, as an enterprise grows, the share of its production destined for international markets tends to expand. Larger enterprises command superior organizational, financial, and managerial resources, allowing them to absorb more readily the fixed sunk costs of foreign market entry. The case of YPF (*Yacimientos Petrolíferos Fiscales*) is particularly telling: following its partial privatization, the corporation mobilized substantial financial and managerial capital to expand foreign sales of petroleum and natural gas. Its scale enabled it to cover the heavy fixed costs of export entry, such as logistical investments, compliance with international standards, and complex cross-border contractual management. This trajectory directly aligns with heterogeneous firm theory, formalized in Melitz's (2003) model, which establishes that only the most productive, large-scale firms can cover the fixed costs of exporting, concentrating foreign trade among a small cadre of large corporations. Such enterprises also benefit from economies of scale that lower unit costs and sharpen international competitiveness. Scale alone, however, does not guarantee export success: across specialized niche industries, product differentiation, process efficiency, and integration into value chains remain equally decisive. Empirical studies document that expanding firm size correlates with greater outward orientation and higher export shares across sectors such as automotive manufacturing, steel production, and retail distribution (Toulan, 1997; Guillén, 2002).

Conversely, small and medium enterprises (SMEs), although frequently motivated to seek foreign outlets due to fierce domestic competition, face substantial entry barriers. Constraints linked to sunk market entry costs, limited resource buffers, and the complexity of foreign regulations impede internationalization. These barriers include high sanitary or technical certification expenses, mandatory product adaptations, and the capital required to build foreign distribution networks. Despite these obstacles, studies such as Albornoz and Ercolani note that small firms that successfully enter international markets can experience substantial productivity gains. Their flatter, more flexible organizational hierarchies can facilitate the internal dissemination of knowledge acquired abroad, although these advantages do not always compensate for high entry barriers.

Human Capital and Workforce Qualification

Workforce educational attainment also plays a central role in an enterprise's international orientation. Research by Toulan and Guillén indicates that firms with a higher proportion of unskilled labor face lower export probabilities, as internationalization requires specialized organizational and cognitive capabilities. Deficiencies in human

capital hinder a firm's ability to process, interpret, and implement complex information regarding foreign markets. Exporting, particularly to advanced economies, requires compliance with technical norms, sanitary standards, and sophisticated commercial procedures that demand qualified personnel. Conversely, firms with a skilled labor force are better equipped to absorb technical, logistical, and commercial knowledge from abroad. This learning capacity enables them not only to meet foreign requirements, but also to upgrade product quality and production workflows, bolstering sustained international competitiveness.

Empirical findings from Barletta, Pereira, and Yoguel, based on a sample of 2,548 Argentine manufacturing SMEs surveyed in 2009, corroborate this central role of human capital: while only 35.3% of firms were categorized as possessing "high" technological capabilities (share of qualified labor), these firms exhibited a substantially higher average export intensity (19.6% of sales among exporting firms). Albornoz and Ercolani further demonstrate that exporters endowed with skilled labor ("Skilled-Labor exporters") capture significant productivity dividends, as human capital enhances an enterprise's absorptive capacity to interpret and leverage knowledge gained through foreign market exposure. Exporting exposes firms to demanding technical, organizational, and competitive environments that generate learning opportunities. However, such opportunities translate into productivity enhancements only when the enterprise commands sufficient human capital to integrate this knowledge into internal operations. Qualified labor thus acts as a key transmission channel for "learning by exporting," facilitating internal knowledge diffusion and process adaptation, explaining the higher productivity gains observed among skill-intensive exporters.

Industry Sector

According to ECLAC (Economic Commission for Latin America and the Caribbean), Argentine industry has experienced an evolution marked by profound structural ruptures. The forced opening between 1976 and 1983, a period during which the military dictatorship imposed abrupt trade liberalization by drastically slashing import tariffs and industrial protections, led to a severe contraction of domestic industrial output. Sectors such as textiles, woodworking, and metallurgy experienced production collapses between 30% and 40%. Manufacturing's share of national GDP contracted from 28% in 1975 to 22% in 1982, and nearly 20% of major manufacturing plants ceased operations. This loss of industrial capacity durably curtailed the manufacturing sector's contribution to national exports. Conversely, agriculture and agribusiness emerged as the central engine of Argentine exports. According to Saini and Lema, grain and oilseed cultivation achieved remarkable productivity gains: in certain agricultural zones, wheat yields advanced by approximately 5% annually, while soybean yields grew between 2.4% and 2.8% per year. These productivity gains were driven by the widespread adoption of technological innovations including direct seeding (no-till farming), genetically modified seed varieties, and modern storage facilities (silobags). Supported by these technical and organizational advances, agribusiness solidified its standing as the most dynamic sector and primary source of Argentine export earnings.

Female Top Management

Within the empirical literature on business internationalization, the role of female executive leadership has attracted increasing scrutiny. Early studies highlighted a lower propensity to export among female-led firms (Westhead et al., 2001; Nissan et al., 2012), a pattern also documented by Alves et al. (2017, 2021) across multi-country samples. However, this empirical relationship is far from uniform. Orser et al. (2010) show that gender discrepancies largely diminish once controlling for firm size and sector, whereas Spence et al. (2011) observe a higher export propensity among female-led startup ventures. Similarly, Karam and Zaki (2021) report a positive effect of female leadership within large enterprises across the Middle East and North Africa. On theoretical grounds, Bannò and Filippi (2024) observe that these mixed findings stem from multiple confounding factors, including the frequent omission of gender-informed theoretical frameworks, the disproportionate concentration of female executives within

structurally less internationalized sectors (selection bias), and gender-differentiated access to credit, international networks, or risk preferences (McClelland et al., 2005; Gundlach and Sammartino, 2019). This literature motivates the inclusion of a female management indicator in our model, although its expected coefficient sign remains theoretically ambiguous.

2.2 External Factors

Geographical Structure of Exports

The spatial distribution of economic activity represents a decisive determinant of trade outcomes. In “Regional Export Performance: First Nature, Agglomeration and Destiny?”, findings highlight an acute geographic concentration of Argentine exports that reflects deep inter-regional structural disparities. The Pampeana region constitutes the economic and export heartland of the nation: in 2005, it generated over 70% of total exports and nearly 80% of manufactured exports, while concentrating 70% of national GDP and 80% of industrial production. This dominance stems largely from superior infrastructure endowments; comparative indices reveal that Pampeana and Patagonia possess the highest infrastructure densities, in sharp contrast to the Northwest (NOA) and Northeast (NEA), which remain the least equipped regions in the country. Furthermore, the Pampeana region’s proximity to primary exit nodes—such as the deep-water ports of Buenos Aires and Rosario, located within 369 km of Córdoba and 716 km of central Santa Fe—confers a substantial logistical cost advantage. Conversely, peripheral regions, each accounting for merely 5% to 7% of national exports, contend with structural remoteness and sparse industrial agglomeration. The literature clearly indicates that Argentina’s export hierarchy is heavily shaped by industrial clustering, infrastructure quality, and logistical connectivity, overwhelmingly favoring the central Pampean zone.

2.3 Investment Factors

Research and Development

Enterprises that invest in research and development exhibit a higher likelihood of exporting as well as greater export intensity. Albornoz and Ercolani demonstrate that innovation-driven productivity gains positively influence export outcomes, with evidence pointing toward a bidirectional relationship: exporting enhances internal capabilities, but the most productive firms are precisely those capable of entering international markets, commanding the cost efficiencies and quality attributes required to cover entry sunk costs and compete abroad.

Schmidt and Trofimenko (2010) confirm this strong association between technological investment and export performance: a 1% increase in expenditures on imported technologies increases exports by 44%, compared to 35% for domestic technological acquisitions. Stated differently, foreign sales prove highly elastic with respect to technological investments: modest shifts in technology spending correspond to disproportionate expansions in overseas sales. This responsiveness arises because advanced technology enhances factor productivity, curbs unit costs, and upgrades product specifications, reinforcing competitiveness in foreign markets. This underlines R&D investment as a fundamental pillar of export success. Relevant expenditures span computational infrastructure, specialized software, patent acquisitions, technology transfers, and formal R&D initiatives.

International Quality Certifications

Across Argentine sectors serving export markets subject to stringent regulatory regimes, international certifications play a decisive role. The fisheries sector, investigated by Pérez-Ramírez et al. (2012) through Marine

Stewardship Council (MSC) certification, offers a compelling illustration. While Argentina accounts for approximately 1% of global maritime catches, it ranks among the leading commercial seafood exporters. At the time of their study, four Argentine fisheries participated in MSC certification, with certified catches ranging between 2,000 and 20,000 metric tons annually depending on the species (anchovy, Patagonian grenadier/hoki, Patagonian scallop).

Securing these international credentials directly enhanced market access to European and North American buyers, where demand for sustainably certified seafood has expanded rapidly. This dynamic underscores that conformity with international standards acts as an essential catalyst for export expansion. It further emphasizes the need to cultivate local capacities in quality auditing, value-added processing, and technical training so that domestic producers can capture a larger share of value-added within the country.

Digitalization

Digital transformation has emerged as a fundamental engine of business modernization and competitive strategy. Calderon-Monge and Ribeiro-Soriano (2024) highlight that digital technologies, notably SMAC tools (social, mobile, analytics, and cloud computing), fundamentally transform how enterprises engage with clients and enter new markets. Digital technologies strengthen international visibility and mitigate information asymmetries (Chan et al., 2022). In marketing, social media platforms and digital channels have become indispensable pathways for connecting with international audiences (Lamberton and Stephen, 2016). An active web presence enhances customer engagement, facilitates reputational word-of-mouth, and drives value co-creation, all of which support corporate internationalization. From a managerial standpoint, digitally mature firms are positioned to seize international opportunities more rapidly (Nasiri et al., 2022). In this respect, maintaining an active corporate website serves as an observable proxy for modernization and outward orientation, reducing market entry costs abroad (Pergelova et al., 2019), particularly for resource-constrained SMEs (Rosenbaum, 2017). Given empirical limitations in capturing complete digital maturity across survey data, we retain corporate website ownership as an observable proxy for digitalization, positing an expected positive relationship with export probability.

Drawing upon the theoretical and empirical literature, several hypotheses are formulated regarding the determinants of export propensity among Argentine firms:

- H1 - Firm Size:** Larger enterprises exhibit a higher probability of exporting due to their superior capacity to absorb fixed entry costs in foreign markets. Hence, firm size is positively associated with export intensity.
- H2 - Human Capital Qualification:** The literature indicates that a more qualified labor force facilitates integration into international markets. The share of skilled employees is positively associated with export propensity.
- H3 - Industry Sector:** Specific sectors, notably manufacturing and agro-industrial activities, are structurally oriented toward international commerce. The industry sector significantly influences the probability of exporting.
- H4 - Geographic Location:** Exports are heavily concentrated in the Pampeana region due to superior infrastructure (ports, transport arteries, logistics), suggesting that firms in well-endowed regions export more frequently.
- H5 - International Certification:** Quality certifications ease market access into demanding regulatory environments. Holding an international certification increases export engagement.
- H6 - R&D Expenditures:** Technological investment is a primary driver of foreign sales. Firms investing in R&D display a higher propensity to export.

- H7 - Digitalization (Website):** Operating a website enhances global visibility and commercial linkages. Website ownership positively affects export probability.
- H8 - Female Top Management:** The literature suggests potential linkages between leadership diversity and innovation. Female top management may positively influence export propensity.

3 Empirical Study

Data Description

The empirical dataset utilized in this research is drawn from the 2017 World Bank Enterprise Survey (WBES) for Argentina, accessible through the official World Bank microdata repository: <https://microdata.worldbank.org/index.php/catalog/3386>.

The WBES is designed to collect firm-level data on the business operating environment across developing and emerging economies, covering access to finance, innovation, governance, and participation in international trade. The survey employs a stratified random sampling methodology to ensure representation of Argentina's formal private sector along three principal strata: economic sector (manufacturing, services, construction, commerce), firm size (small, medium, large), and geographic region (Buenos Aires, Córdoba, Santa Fe, Mendoza, Tucumán, etc.). The raw questionnaire comprises 313 variables, with a cross-sectional sample of approximately 1,000 registered enterprises employing at least five full-time workers. Survey interviews were administered face-to-face with business owners or senior executives throughout 2017.

To prepare the dataset for analysis, several data processing steps were executed. First, variable filtering was conducted: among the hundreds of survey metrics available, only variables pertinent to export capacity and theoretical determinants were retained. These capture structural characteristics (size, sector, region), technological and organizational capabilities (website presence, international certification, workforce qualification, R&D spending), and the primary dependent variable of export activity. Variables were systematically renamed for consistency. Binary qualitative indicators were converted into dummy variables (0/1). Following data cleaning and listwise deletion of incomplete observations, the working sample consists of 991 enterprises and 9 primary analytical variables. The subsequent section outlines the selected variables.

Table 1: Variable Structure of the Empirical Dataset

Variable	Description	Data Type
Export	Percentage of total annual sales derived from direct exports.	Quantitative
Size (Taille)	Enterprise size category (small, medium, large).	Qualitative
Region	Geographic location of the enterprise (Buenos Aires, Mendoza, Córdoba...).	Qualitative
Industry Sector	Main economic sector of the enterprise.	Qualitative
Website	Indicates whether the firm operates an active website (1 = Yes, 0 = No).	Qualitative (binary dummy)
Qualification	Proportion of qualified/skilled workers within the enterprise.	Quantitative
International Certif.	Indicates whether the firm holds an internationally recognized certification.	Qualitative (binary dummy)
R&D Expenditure	Indicates whether the firm engaged in R&D expenditures during the fiscal year.	Qualitative (binary dummy)
Female Executive	Indicates whether the enterprise has a female top manager.	Qualitative (binary dummy)

Variables Related to Exporting Activity

The core dependent variable considered is the percentage of sales generated through direct exports (*Export*). This metric measures an enterprise's export intensity: a higher percentage reflects deeper integration into international markets. Utilizing a continuous share rather than a strict binary dummy enables a finer evaluation of export engagement, distinguishing firms that export marginally from those that allocate the majority of their production to international markets.

Structural Determinants

Firm size is a primary structural determinant. Larger corporations typically command greater organizational and financial reserves, allowing them to support larger export operations and cover fixed entry expenses. Industry sector reflects the structural export orientation of an enterprise's underlying product line. Sectors such as manufacturing, agro-industrial processing, and precision equipment possess an intrinsic outward orientation compared to domestic-focused services. Finally, geographic region of operation exerts a strong spatial influence. Enterprises located in Buenos Aires or nearby coastal hubs benefit from immediate proximity to container ports and industrial clusters, securing lower transport costs and enhanced access to foreign buyers.

Organizational and Technological Capabilities

Firm-internal characteristics play an essential role in determining export volume, capturing innovation, operational modernization, and professionalization.

Workforce qualification reflects internal human capital endowments. Firms staffed by skilled personnel possess productivity advantages that enable them to satisfy stringent technical specifications demanded abroad. Operating an active corporate website indicates digital modernization and international transparency, facilitating contact with foreign partners and enhancing global visibility.

Holding an international quality certification signals adherence to rigorous production standards, facilitating integration into global value chains. Expenditures on research and development capture the internal commitment to process and product innovation, which is critical for competing in contestable foreign markets. Finally, female executive leadership provides insight into corporate managerial diversity and organizational governance.

Statistical Methodology and Data Processing

Statistical analyses were executed using R Studio. All statistical hypothesis tests reported throughout this paper were evaluated against a significance threshold of $\alpha = 5\%$. To examine distribution patterns and descriptive trends across the sample, bar charts and histograms were produced.

Descriptive Analysis

Within the full sample of 991 enterprises, 15.3% engage in direct exports, whereas 84.7% sell exclusively to the domestic market. The following section examines the sample distribution across the analyzed variables.

Descriptive statistics for variables among exporting enterprises:

Figure 1: Distribution of Enterprise Size

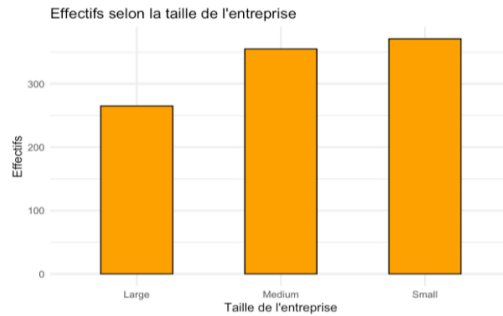
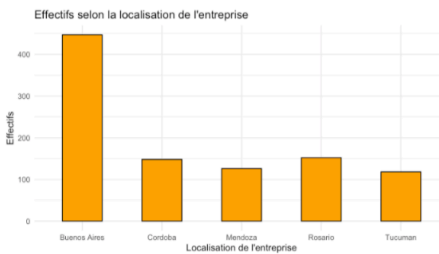


Figure 1 illustrates a relatively balanced distribution across firm size categories, though small and medium-sized enterprises constitute the majority. This distribution motivates testing whether smaller firm size acts as a significant constraint on international market entry.

Figure 2: Distribution of Enterprise Geographic Location



Regarding spatial distribution, a substantial majority of enterprises are concentrated in the Greater Buenos Aires region, reflecting national economic centralization and highlighting potential infrastructure bottlenecks for peripheral provinces.

Figure 3: Distribution of Corporate Website Ownership

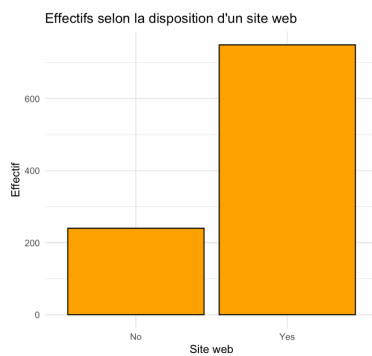


Figure 3 indicates that an overwhelming majority of sampled enterprises maintain an active corporate website, suggesting that basic digital connectivity has become a baseline standard for commercial operations.

Figure 4: Distribution Across Industry Sectors

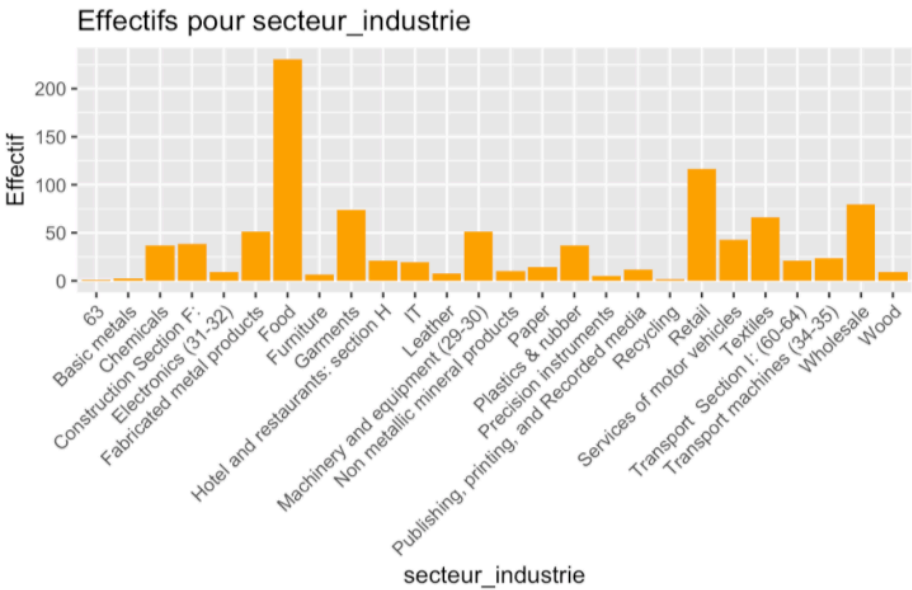


Figure 4 highlights the strong presence of the agro-food industry (*Food*) within the sample, alongside broad diversity across manufacturing and trade activities (*Retail*, *Wholesale*). This sectoral heterogeneity underscores the necessity of industry-level controls to determine whether specific lines of activity facilitate or hinder access to foreign markets.

Bivariate Association Analysis

To develop initial empirical insights into export determinants, we conduct bivariate analyses across quantitative, binary, and qualitative variables.

Figure 5: Correlation Matrix

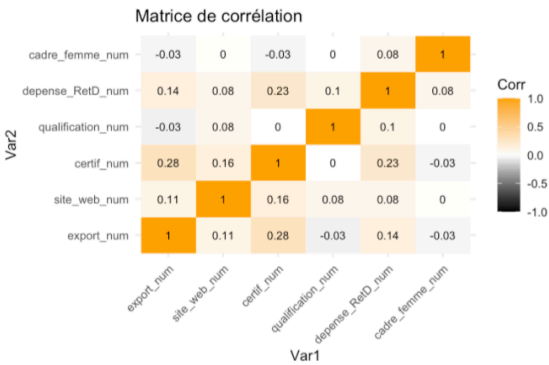


Figure 5 displays the correlation matrix across primary model covariates. Estimated correlation coefficients remain consistently modest, all below 0.30, mitigating concerns over severe multicollinearity and ensuring stability in subsequent econometric estimations.

Figure 6: Relationship Between R&D Expenditure and Mean Export Intensity

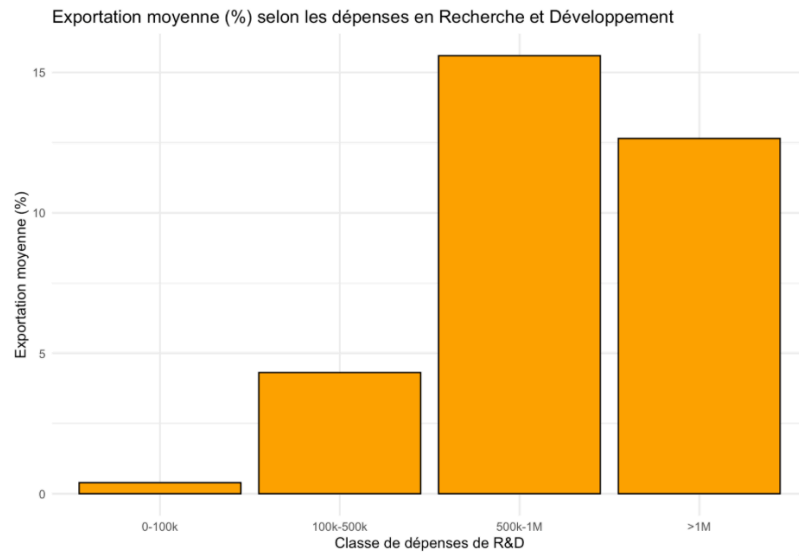


Figure 6 illustrates a positive correlation between R&D investment and export intensity. Higher R&D outlays coincide with a larger share of revenue earned abroad, supporting the notion that technological innovation functions as a key non-price competitive advantage.

Figure 7: Mean Export Intensity by Firm Size



Figure 7 indicates that the average export share increases monotonically with enterprise size. This pattern is consistent with international trade literature positing that larger firms command superior resources to absorb the fixed costs of entering foreign markets.

Figure 8: Mean Export Intensity by Geographic Region

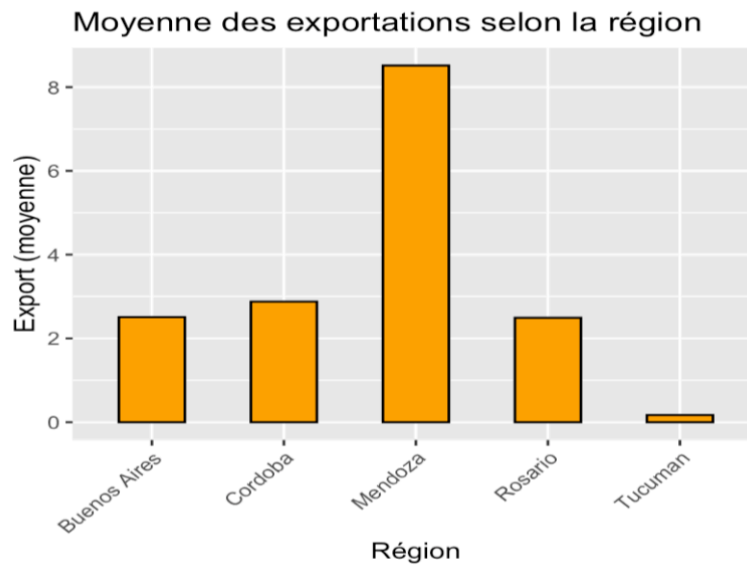


Figure 8 highlights marked geographic disparities in export intensity. The Mendoza region displays the highest mean export share, whereas Tucumán records the lowest, pointing to a pronounced spatial dimension in firms' international orientation.

Figure 9: Mean Export Intensity by International Certification Status



Figure 9 reveals a pronounced difference in mean export intensity between certified and non-certified enterprises. Holding an international quality certification appears to act as a crucial signal of quality, strongly associated with deeper integration into foreign markets.

Figure 10: Mean Export Intensity by Industry Sector

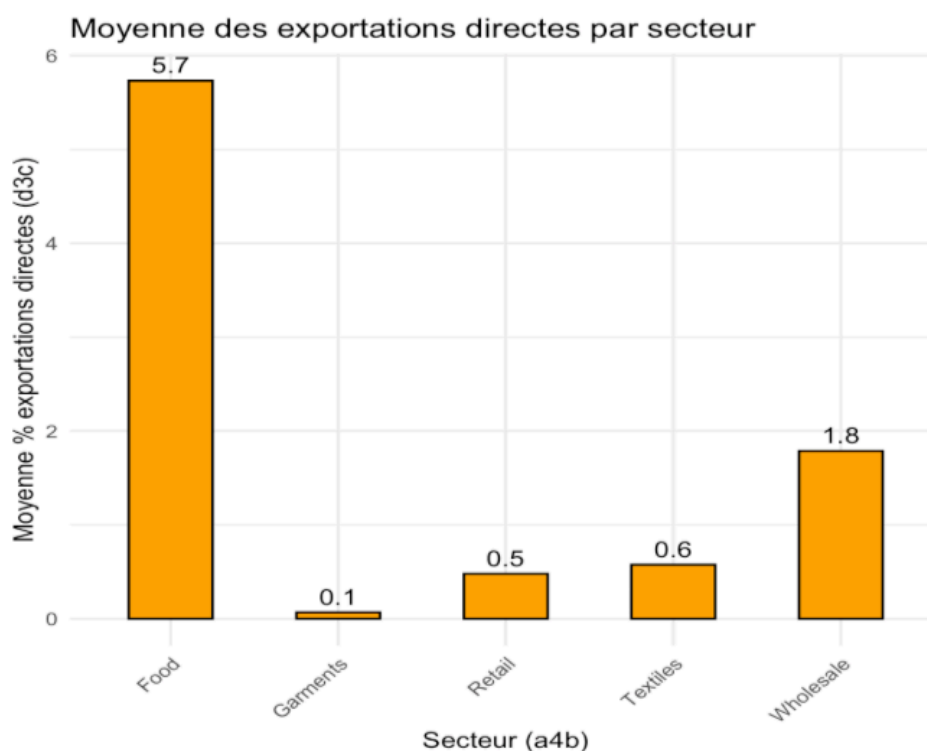


Figure 10 confirms the predominance of the agro-food industry (*Food*) in terms of direct export intensity, averaging 5.7%, well above other sampled sectors. This outcome underlines Argentina's comparative advantage in agro-industrial production, whereas retail and textile branches exhibit far lower export involvement.

ANOVA TESTS

To assess bivariate associations between the continuous export intensity metric and multi-categorical covariates (e.g., firm size, region, industry sector), we perform Analysis of Variance (ANOVA). ANOVA tests whether the mean of a continuous metric (export percentage) differs significantly across groups defined by a categorical variable.

The objective is to determine whether at least one group mean deviates significantly from the others. ANOVA is especially suited for simultaneously comparing more than two group means.

The underlying statistical hypotheses are:

- **Null Hypothesis (H_0):** Mean export intensities are identical across all categories.
- **Alternative Hypothesis (H_1):** At least one category mean differs significantly from the others.

When the resulting p-value falls below the significance threshold of $\alpha = 0.05$, the null hypothesis is rejected, indicating statistically significant differences in mean export intensity across groups.

Table 2: Exact ANOVA Test Results

Variable	p-value	F value
Size (Taille)	0.0104	4.591
Region	2.22e-07	9.301
Industry Sector	1.67e-05	2.684

The ANOVA results confirm that Firm Size, Geographic Region, and Industry Sector each exert a statistically significant influence on export intensity, with each variable yielding a p-value below 5 %. Geographic Region exhibits the largest test statistic ($F = 9.301$), indicating that spatial location provides the strongest source of cross-group variation in export intensity.

STUDENT'S T-TESTS

The purpose of the two-sample Student's t-test is to evaluate whether two independent groups display statistically different mean export intensities. The *t-test* is appropriate for evaluating binary categorical covariates.

The hypotheses tested are:

- **Null Hypothesis (H_0):** Mean export intensities are identical between the two groups.
- **Alternative Hypothesis (H_1):** Mean export intensities differ significantly between the two groups.

If the calculated p-value is below the threshold $\alpha = 0.05$, the null hypothesis is rejected, providing evidence of a statistically significant difference between group means.

Table 3: Two-Sample T-Test Results

Variable 0 = No/Does not hold 1 = Yes/Holds	p-value	95% CI	Mean Exports (Group 0)	Mean Exports (Group 1)
International Certification	1.666e-08	[-10.40; -5.16]	1.33%	9.11%
Female Top Executive	0.3011	[-1.14; 3.64]	3.15%	1.90%
Website	3.509e-06	[-4.28; -1.75]	0.77%	3.78%

The t-tests demonstrate that possessing an International Certification and maintaining a corporate Website are associated with significantly higher export intensity, with equipped enterprises displaying markedly superior mean export shares (9.11% vs. 1.33%, and 3.78% vs. 0.77%, respectively).

Conversely, the Female Top Executive dummy displays no statistically significant association with export intensity, as its p-value (0.3011) remains well above the 5% threshold.

Summary of Test Results

This section summarizes the primary bivariate findings from the 2017 WBES dataset. Within the sample, only 15.3% of enterprises engage in exporting, corroborating the modest overall trade participation of Argentina's productive base.

Structural Characteristics

Firm size emerges as a key structural factor: the ANOVA ($p = 0.0104$) demonstrates that large enterprises export a significantly higher proportion of output than small enterprises. The Fisher statistic ($F = 4.591$) confirms that the between-group variance accounted for by firm size significantly exceeds residual within-group variance. As $p < 0.05$, we reject H_0 in favor of H_1 , validating the significance of enterprise scale.

Likewise, geographic region plays a decisive role ($p = 2.22 \times 10^{-7}$): firms located in infrastructure-rich provinces (Buenos Aires, Mendoza) register higher export intensity. Economic sector similarly influences export orientation ($p = 1.67 \times 10^{-5}$), with stronger engagement concentrated across agro-food and manufacturing branches.

Organizational and Technological Capabilities

Among internal firm capabilities, international certification displays a particularly strong relationship: t-test results indicate that certified enterprises export, on average, nearly seven times more than non-certified firms (Figure 9), with mean export intensity of 9.11% for certified firms versus 1.33% for non-certified firms, supporting Hypothesis 5.

The 95% confidence interval for the mean difference $[-10.40; -5.16]$ further confirms the statistical robustness of this gap. R&D expenditures and website ownership also exhibit positive bivariate correlations with export activity, suggesting that technological capabilities and digital readiness bolster trade engagement. Labor force qualification likewise appears positively associated with export volume.

Managerial Structure

By contrast, the presence of a female top executive is not statistically significant ($p = 0.3011$), indicating that gender composition in top management is not systematically associated with export intensity within this sample.

4 Econometric Analysis

Model Specification

Within the binary Probit framework, the probability that enterprise i participates in exporting is formulated as follows:

$$P(\text{Exportdummy}_i = 1|X) = \Phi\left(\beta_0 + \beta_1\text{Size} + \beta_2\text{Region} + \beta_3\text{Qualification} + \beta_4\text{Sector} + \beta_5\text{Certif_internationale} + \beta_6\text{Site_web} + \beta_7\text{Cadre_femme} + \beta_8\text{R\&D}\right) \quad (1)$$

where $\Phi(\cdot)$ denotes the cumulative distribution function (CDF) of the standard normal distribution, bounding predicted probabilities within the unit interval $[0, 1]$. Parameter vectors β are estimated via maximum likelihood estimation (MLE).

Several survey variables required recoding prior to model estimation. Categorical questions regarding international certifications (*certif_internationale*), website ownership (*site_web*), and female top management (*cadre_femme*) were converted into binary dummies (0/1). The R&D variable (*a_depense_RetD*) was likewise encoded as 1 if the enterprise incurred research expenditures during the reference year, and 0 otherwise.

Additionally, the detailed industry variable (*secteur_industrie*), originally spanning twenty disaggregated codes, was aggregated into eight homogeneous categories to ensure adequate cell counts and improve estimation precision: agro-food (*Food*), intermediate goods (*Plastics, Paper, Wood*), consumer goods (*Furniture, Leather, Textiles, Apparel*), heavy industry (*Basic Metals, Non-Metallic Minerals, Chemicals*), machinery and equipment (*Machinery, Transport Equipment, Electronics, Precision Instruments*), metal fabrication (*Fabricated Metal Products*), wholesale and retail trade (*Retail, Wholesale, Automotive Repair*), and commercial services (*IT, Hospitality, Publishing, Transport, Recycling, Construction*).

Enterprise size (small, medium, large) and labor qualification (low, medium, high) were structured as three-level ordinal factors, while geographic region was grouped into five primary administrative areas (Buenos Aires, Córdoba, Mendoza, Rosario, Tucumán).

Goodness of Fit

Overall, the estimated Probit specification demonstrates solid explanatory power. Null deviance equals 848.68, which drops to a residual deviance of 614.28 upon incorporating the full vector of explanatory covariates. This reduction of 234.40 points confirms that the included explanatory variables contribute meaningful information, improving the model's capacity to predict export participation.

The Akaike Information Criterion (AIC) for the estimated model is 654.28. By penalizing model complexity against likelihood improvements, the AIC confirms an appropriate balance between goodness-of-fit and statistical parsimony.

Reference Category Profile

The baseline reference category corresponds to a **large enterprise located in Buenos Aires, operating in the agro-food industry, and staffed by a highly qualified workforce**. Estimated coefficients consequently

capture the directional change in the latent export propensity relative to this reference baseline, *ceteris paribus*.

Table 4: Probit Model Estimation Results

Variable	Coefficient	Std. Error	z value	Pr(> z)	Sig.	AME
(Intercept)	-1.546	0.242	-6.378	1.80e-10	***	
tailleMedium	-0.121	0.142	-0.853	0.394		-0.0218
tailleSmall	-0.213	0.157	-2.154	0.031	*	-0.0372
regionCordoba	-0.040	0.163	-0.246	0.806		-0.0074
regionMendoza	0.104	0.170	0.615	0.538		0.0203
regionRosario	-0.100	0.167	-0.602	0.547		-0.0181
regionTucuman	-1.636	0.497	-3.289	0.001	**	-0.1529
qualification_Low	-0.234	0.128	-1.825	0.068	.	-0.0403
qualification_medium	-0.142	0.166	-0.852	0.394		-0.0252
Biens_consommation	-0.314	0.194	-1.615	0.106		-0.0532
Biens_intermediaires	0.444	0.218	2.036	0.042	*	0.1001
Commerce	-0.452	0.192	-2.360	0.018	*	-0.0722
Industrie_lourde	0.385	0.239	1.608	0.108		0.0852
Machines_equipements	0.398	0.192	2.075	0.038	*	0.0884
Services	-0.517	0.237	-2.183	0.029	*	-0.0801
Transformation_metal	0.021	0.263	0.081	0.936		0.0041
certif_inter_dummy	0.818	0.130	6.302	2.94e-10	***	0.1402
site_web_dummy	0.623	0.190	3.279	0.001	**	0.1067
cadre_femme_dummy	-0.415	0.246	-1.690	0.091	.	-0.0711
a_depense_RetD	0.633	0.145	4.376	1.21e-05	***	0.1084

Significance codes: < 0.001 '***'; < 0.01 '**'; < 0.05 '*'; < 0.1 '.'

Note: AME = Average Marginal Effects.

5 Econometric Estimation Findings

This section examines the Probit estimation results across individual covariates, identifying the structural, geographic, and organizational factors that determined the probability of Argentine firms exporting in 2017.

1. Firm Size

Estimation results for firm size provide partial support for Hypothesis H_1 . While the medium-sized category (*tailleMedium*: $\beta = -0.121$; $p = 0.394$) exhibits no statistically significant difference relative to large corporations at the 5% level, small firms (*tailleSmall*) display a negative and statistically significant coefficient ($\beta = -0.213$; $p = 0.031$). The corresponding average marginal effect ($AME = -0.0372$) indicates that small firms have, *ceteris paribus*, a 3.7 percentage point lower probability of exporting compared to large enterprises.

This finding aligns with heterogeneous firm theory (Melitz, 2003), which posits that smaller firms struggle to overcome the fixed entry costs associated with foreign market entry. Conversely, the absence of a statistically significant divergence between medium and large firms suggests that once organizational capabilities—such as quality certifications, R&D activity, and digital infrastructure—are controlled for, scale disadvantages are partially attenuated. Consequently, Hypothesis H_1 is partially validated.

2. Geographic Location

Geographic location represents one of the most substantial determinants of export probability. The regional indicators for Córdoba, Mendoza, and Rosario do not differ significantly from Buenos Aires, pointing to comparable export conditions across these central industrial provinces.

Conversely, the peripheral province of Tucumán displays a large, negative, and statistically significant coefficient at the 1% level ($\beta = -1.636$; $p = 0.001$). The average marginal effect reveals a substantial geographic penalty: operating out of Tucumán reduces the probability of exporting by 15.3 percentage points ($AME = -0.1529$) relative to an otherwise identical firm in Greater Buenos Aires.

This result reflects transport cost differentials and logistics bottlenecks. Empirical evidence in Granato (2008) emphasizes that Argentine exports remain heavily concentrated in the central Pampeana region, which accounts for over 70% of total exports and 80% of manufactured exports. This region benefits from dense industrial clustering, modern transport corridors, and direct access to primary maritime terminals. Peripheral regions such as Tucumán face structural remoteness, higher transit times, and greater freight expenses, reducing the price competitiveness of regional producers. These findings confirm Hypothesis H_4 .

3. Human Capital Qualification

The qualification level of the labor force displays a marginally significant relationship with export probability. The coefficient for low workforce qualification ($\beta = -0.234$; $p = 0.068$) is negative and significant at the 10% level. Marginal effect calculations indicate that relying primarily on low-skilled labor reduces the export probability by approximately 4 percentage points ($AME = -0.0403$) compared to firms with highly qualified personnel. Consistent with Toulan and Guillén (1997), a skilled workforce enhances an enterprise's absorptive capacity to interpret technical standards, fulfill export documentation, and navigate foreign regulatory environments. These results provide partial support for Hypothesis H_2 .

4. Industry Sector

Industry sector plays a prominent role in determining export participation. Firms in intermediate goods ($\beta = 0.444$; $p = 0.042$) and machinery and equipment ($\beta = 0.398$; $p = 0.038$) display positive, statistically significant coefficients at the 5% level. In terms of average marginal effects, operating in intermediate goods or machinery increases the probability of exporting by 10.0 percentage points ($AME = 0.1001$) and 8.8 percentage points ($AME = 0.0884$), respectively, relative to the baseline agro-food sector.

This pattern reflects modern global value chain fragmentation: intermediate inputs and specialized machinery are actively traded across international borders as production stages are dispersed internationally. Furthermore, producers of machinery benefit from global demand for industrial capital goods.

In contrast, commerce ($\beta = -0.452$; $p = 0.018$) and commercial services ($\beta = -0.517$; $p = 0.029$) exhibit negative, statistically significant coefficients, corresponding to marginal reductions of 7.2 percentage points ($AME = -0.0722$) and 8.0 percentage points ($AME = -0.0801$), respectively. Retail and wholesale enterprises act primarily as local intermediaries rather than producers of tradable output, while many services face high non-tradability due to face-to-face requirements. These structural dynamics restrict cross-border trade, validating Hypothesis H_3 .

5. International Certifications

Holding an international quality certification is among the strongest predictors of export activity, with a positive coefficient significant at the 0.1% level ($\beta = 0.818$; $p = 2.94 \times 10^{-10}$). The estimated average marginal effect indicates that possessing an international certification raises the probability of exporting by 14.0 percentage points ($AME = 0.1402$), all else equal.

This impact operates through two key mechanisms. First, certifications verify compliance with global safety, environmental, and technical benchmarks, lowering regulatory barriers in destination markets. Second, certifications address information asymmetries by serving as credible signals of product reliability and managerial competence to foreign buyers (Pérez-Ramírez et al., 2012). These results corroborate Hypothesis H_5 .

6. Digitalization: Corporate Website

Operating a corporate website is positively and significantly associated with export propensity at the 1% level ($\beta = 0.623$; $p = 0.001$). The marginal effect indicates that maintaining a website increases the probability of exporting by 10.7 percentage points ($AME = 0.1067$). This result supports the framework of Calderon-Monge and Ribeiro-Soriano (2024), who highlight that digital presence enhances global visibility and reduces search and matching frictions in international commerce, and aligns with Pergelova et al. (2019) regarding the role of digital tools in lowering foreign market entry costs for SMEs. A functional digital presence provides an accessible, low-cost channel for foreign clients to discover domestic suppliers, confirming Hypothesis H_7 .

7. Research and Development Expenditures

R&D spending is positively and significantly linked to export probability at the 0.1% level ($\beta = 0.633$; $p = 1.21 \times 10^{-5}$). Evaluating the average marginal effect indicates that committing funds to R&D increases the probability of exporting by 10.8 percentage points ($AME = 0.1084$).

R&D activity serves as a primary engine of international competitiveness. By innovating, firms adapt products to foreign requirements, improve quality, and enhance operational efficiency, differentiating themselves in competitive export markets. These results echo Schmidt and Trofimenko (2010) regarding the technological sensitivity of Argentine exports and support the bidirectional relationship between innovation and trade identified by Alborno and Ercolani (2007). These findings confirm Hypothesis H_6 .

8. Female Top Management

The inclusion of the female executive dummy (`cadre_femme`) allows for an empirical evaluation of executive gender on export behavior. The variable exhibits a negative coefficient ($\beta = -0.415$) that does not achieve statistical significance at the conventional 5% threshold ($p = 0.091$). A larger sample would be required to draw definitive inferences.

At the 10% level, however, female top management is associated with a lower export probability, corresponding to an average marginal effect of -7.1 percentage points ($AME = -0.0711$). This finding aligns with Bannò and Filippi (2024), who document that gender effects in internationalization often stem from selection biases, such as female executives being more prevalent in domestically focused service sectors or facing unequal access to commercial export networks. Hypothesis H_8 is therefore not supported.

Synthesis of Econometric Findings

The econometric analysis reveals several principal determinants of export behavior among Argentine firms. Investments in R&D, international quality certifications, corporate web presence, and specialization in capital or intermediate goods significantly raise the likelihood of export participation.

Conversely, geographic remoteness (Tucumán), low workforce qualification, and operation within retail trade or local services represent substantial constraints on international market engagement. Firm size matters primarily at the lower bound: while small firms experience a modest export penalty ($AME = -0.037$), medium enterprises do not differ significantly from large corporations once internal capabilities are accounted for. These findings support heterogeneous firm models, confirming that innovation, quality compliance, and productivity are central to overcoming foreign market entry costs.

6 Predictive Profiling

Following the econometric estimation, several covariates emerged as statistically significant drivers of export participation. Using these empirical findings, we construct characteristic profiles comparing enterprises across different predicted probability thresholds.

Using the fitted Probit specification, we compute the fitted probability of exporting for every enterprise in the working sample. The tables below outline the structural and operational characteristics distinguishing firms with a high predicted probability of exporting ($p > 80\%$) from those with a low predicted probability ($p < 20\%$).

These profiles illustrate the operational disparities between globally engaged firms and domestically restricted enterprises, spanning digital connectivity, certification status, R&D investments, workforce training, and industrial sector.

Table 5: Profile of Firms with High Export Propensity ($p > 80\%$)

Variable	Proportion / Distribution
Corporate Website	Yes: 100% No: 0%
International Certification	Yes: 100% No: 0%
R&D Expenditures	Yes: 89.66% No: 10.34%
Female Top Executive	Yes: 3.45% No: 96.55%
Workforce Qualification	High Qualification: 72.41% Low Qualification: 20.69% Medium Qualification: 6.90%
Industry Sector	Machinery & Equipment: 41.38% Heavy Industry: 20.69% Intermediate Goods: 17.24%

Table 6: Profile of Firms with Low Export Propensity ($p < 20\%$)

Variable	Proportion / Distribution
Corporate Website	Yes: 68.39% No: 31.61%
International Certification	Yes: 6.27% No: 93.73%
R&D Expenditures	Yes: 6.54% No: 93.46%
Female Top Executive	Yes: 9.95% No: 90.05%
Workforce Qualification	High Qualification: 35.83% Medium Qualification: 13.08% Low Qualification: 51.09%
Industry Sector	Wholesale & Retail Trade: 30.93% Agro-Food: 24.25% Consumer Goods: 18.26%

Comparative Analysis of Enterprise Profiles

Comparing these groupings reveals two distinct enterprise profiles. The representative firm with a high probability of exporting ($p > 80\%$) universally maintains an active website and holds an international quality certification (100% across both dimensions). The vast majority (89.66%) invest continuously in research and development. Furthermore, these firms rely predominantly on skilled labor (72.41% in the high-qualification tier) and are concentrated in advanced manufacturing branches, notably machinery and equipment (41.38%), heavy industry (20.69%), and intermediate industrial goods (17.24%). Female top executive representation within this subgroup remains limited (3.45%).

Conversely, enterprises with a low export propensity ($p < 20\%$) exhibit a markedly different operational profile. Although approximately two-thirds maintain a basic website (68.39%), only 6.27% hold an international certification, and fewer than 7% invest in R&D. Their human capital profile is comparatively lower, with over half the workforce (51.09%) categorized as low-skilled. These enterprises are predominantly situated in domestically focused activities, such as retail and wholesale trade (30.93%), local agro-food processing (24.25%), and standard consumer goods (18.26%).

Finally, female executive management remains low across both groups and does not emerge as a primary differentiating characteristic. Overall, low-propensity firms present greater operational heterogeneity, whereas high-propensity exporters display consistent capabilities in international certification, active R&D commitment, and skilled workforce endowments.

7 Public Policy Recommendations

The econometric findings indicate that an Argentine firm’s internationalization depends not merely on physical scale, but fundamentally on innovation capacity, quality compliance, and geographic connectivity. To support export diversification and expand foreign currency revenues, several targeted policy interventions can be formulated.

To support quality certification and digital adoption, reducing information asymmetries in foreign markets is essential. Uncertainties regarding supplier quality and delivery reliability often deter foreign buyers. Because international quality certifications increase export probability by 14.0 percentage points, and website ownership increases it by 10.7 percentage points, public policy should target these capability gaps. The state could introduce dedicated co-financing grants or tax credits to help SMEs absorb the fixed costs of international audits and quality standards (e.g., ISO, sanitary, and environmental certifications). Concurrently, a structured digital transition program—such as an “Export Digital Voucher”—could assist SMEs with technical and financial resources to deploy multilingual e-commerce platforms and international digital marketing tools.

Additionally, public support should foster industrial innovation within sectors showing proven export potential. In line with heterogeneous firm theory, productivity and innovation capabilities are prerequisites for absorbing foreign entry costs. Our findings show that R&D spending increases export probability by 10.8 percentage points, with intermediate goods and machinery producers benefiting from positive sectoral premiums of 10.0 and 8.8 percentage points, respectively. To build on this momentum, authorities should expand R&D tax credits tailored to export-oriented product adaptations and technological upgrades. This should be complemented by targeted export promotion initiatives (trade fairs, bilateral commercial missions) directed toward competitive capital and intermediate goods manufacturing.

From a spatial perspective, addressing regional infrastructure disparities is critical. Being located in Tucumán incurs a 15.3 percentage point export penalty relative to Greater Buenos Aires. Remoteness from central maritime hubs imposes heavy freight costs, longer transit times, and logistics uncertainty. Remedying this disadvantage requires strategic infrastructure investments to connect the Argentine Northwest (NOA) with central ports, particularly through the modernization of regional freight rail corridors (e.g., the Belgrano Cargas line). In parallel, establishing inland dry ports and specialized customs hubs in peripheral provinces like Tucumán would streamline regulatory clearance, aggregate cargo volumes, and lower unit shipping costs for regional producers.

Finally, policy should address firm size constraints and support human capital formation. Small firms face a 3.7 percentage point structural disadvantage, while low workforce qualification reduces export probability by 4.0 percentage points. Because internationalization demands scale and specialized trade skills, public agencies should encourage export consortia. By pooling resources across SMEs within shared industrial clusters, consortia enable smaller firms to co-finance fixed costs such as foreign market research, legal services, and overseas distribution channels. Simultaneously, the state should collaborate with commercial chambers and universities to deliver practical trade training programs. Specialized coursework covering customs management, international logistics, and trade finance will equip managers with the tools needed to operate in global markets.

Beyond these microeconomic levers, public interventions require a stable macroeconomic environment to achieve lasting impact. Chronic currency volatility, high inflation, and institutional uncertainty represent overarching barriers to business planning. A certified, technologically advanced firm will struggle to sustain an export strategy if it cannot price inputs reliably, access foreign exchange for critical imports, or plan medium-term capital outlays. Microeconomic trade measures must therefore be paired with credible macroeconomic stabilization. In the short term, subsidized exchange rate hedging mechanisms could help shield SME exporters from sudden currency shocks. Over the medium term, anchoring monetary stability, reinforcing central bank credibility, and maintaining fiscal discipline remain necessary prerequisites for Argentine firms to compete durably in international markets.

8 Conclusion

This study examined the determinants of Argentine firms' participation in international trade in 2017 using firm-level data from the World Bank Enterprise Survey. Applying statistical testing and binary Probit modeling, it identified the structural, organizational, technological, and geographic characteristics that shape export propensity. The empirical findings demonstrate that international market participation is highly uneven and driven by specific internal and external capabilities. Among the most influential determinants, international quality certifications and R&D expenditures stand out: holding an international certification increases the probability of exporting by 14.0 percentage points, while investing in R&D raises it by approximately 11 percentage points. These results indicate that export competitiveness relies heavily on technological upgrading and verified adherence to international quality standards rather than price margins alone. Maintaining an active corporate website, serving as an observable indicator of digital readiness, also exerts a statistically significant positive effect, highlighting the role of digital connectivity in foreign market access. On structural dimensions, firm scale plays a qualified role: while small enterprises face a statistically significant disadvantage relative to large corporations, medium-sized firms do not display significant differences. This nuances strict interpretations of Melitz's (2003) model, suggesting that strong organizational capabilities—especially quality certifications and R&D investments—can help mitigate size disadvantages. Sectoral affiliation is also influential: producers of intermediate goods and specialized machinery exhibit higher export propensities, whereas retail commerce and services remain primarily oriented toward the domestic market. Spatially, the analysis reveals persistent regional disparities. Operating out of the peripheral province of Tucumán reduces export probability by more than 15 percentage points relative to Buenos Aires, reflecting logistical frictions and distance from major maritime terminals. These findings illustrate how spatial fragmentation continues to constrain regional productive sectors, regardless of their commercial potential. These results must be viewed within the context of Argentina's macroeconomic environment. The year 2017 represented a period of trade liberalization under the Macri administration, yet structural challenges—including inflation, currency volatility, and capital constraints—remained prominent. In this setting, the findings highlight that internal capabilities can help firms overcome external barriers to foreign trade. Innovation capacity, quality compliance, and digital connectivity emerge as actionable drivers of trade competitiveness across economic cycles.

Ultimately, this study contributes to the empirical literature on microeconomic trade determinants in emerging economies facing structural volatility. It suggests that expanding Argentina's export capacity requires not merely macroeconomic currency adjustments, but sustained institutional support for technological upgrading, quality certification, and freight infrastructure in peripheral regions.

Note on the Use of Artificial Intelligence

In the interest of academic transparency, this report notes the supplementary use of generative artificial intelligence tools. These tools served as technical support for data preprocessing, variable selection, drafting assistance, and L^AT_EX document formatting.

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